MINISTÉRIO DA CIÊNCIA E TECNOLOGIA
INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS

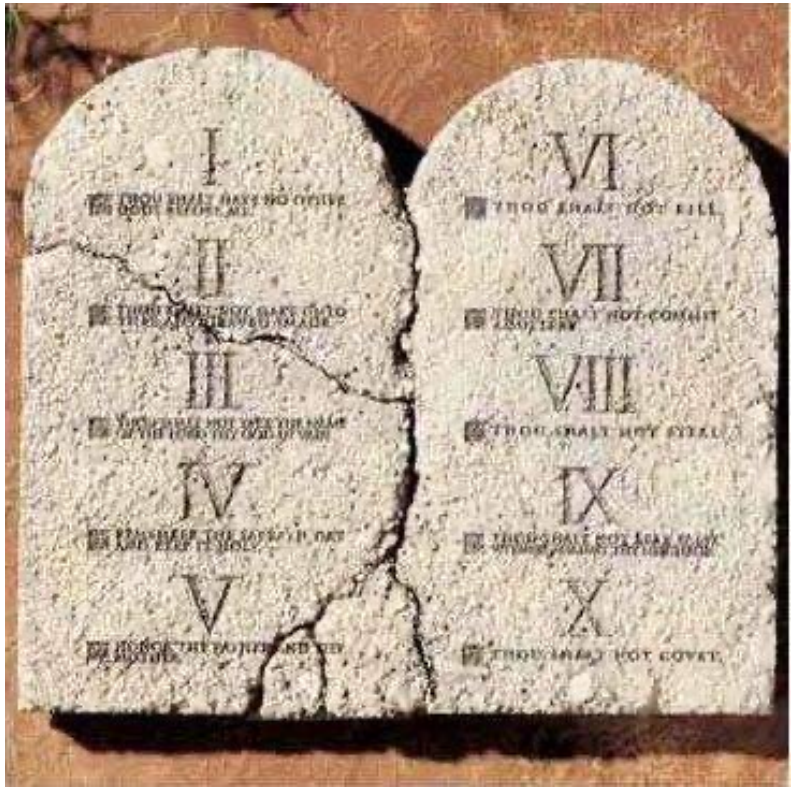
Scientific Research in Practice

Gilberto Câmara

INPE

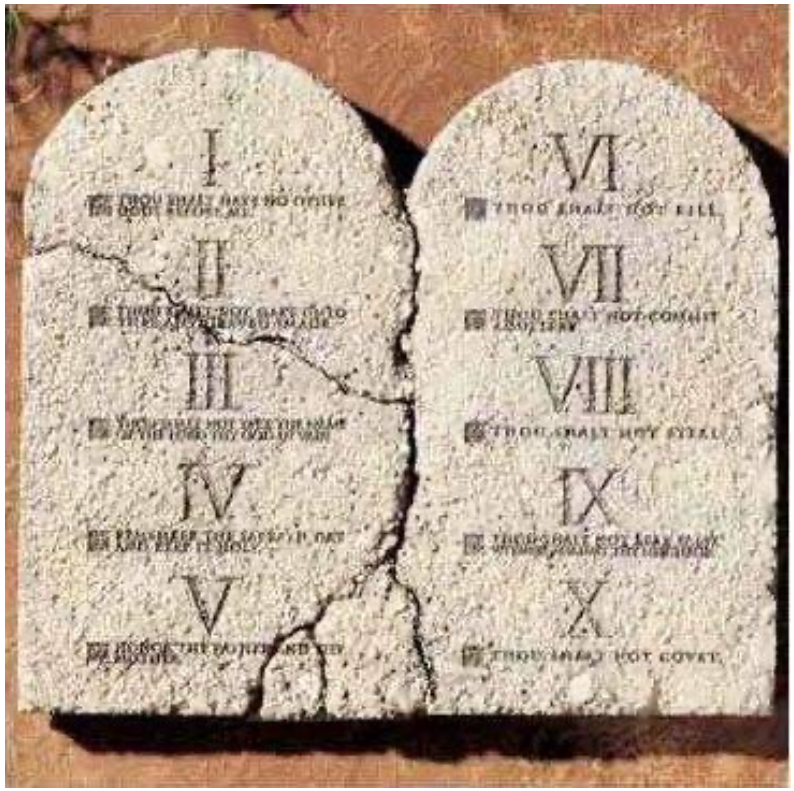
Licensed as Creative Commons – By Attribution – Share Alike – Non-commercial

Ethics: a moral question?



“Thou shall not kill”

Ethics: a moral issue?



"Thou shalt not kill."

What if you needed to kill to escape Auschwitz?

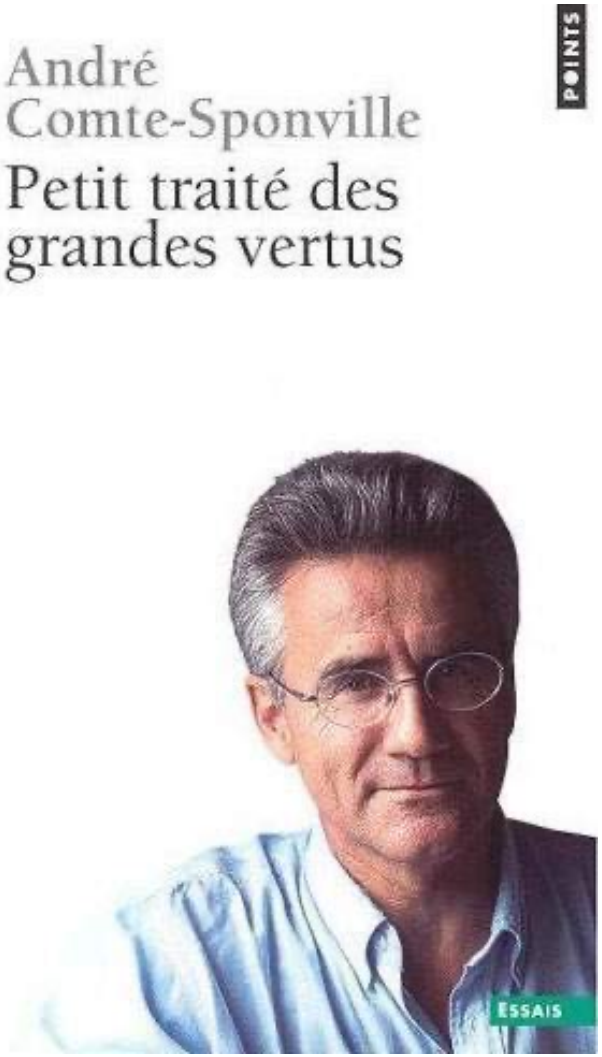
Ethics of Max Weber (1864-1920)

Ethics of conviction: set of norms and values that guide behavior in one's private sphere (moral law)

Ethics of responsibility: set of norms and values that guide decision-making in public life (law of effectiveness)

Ethics in Weber: application of pragmatic (realist) reason

Contemporary Ethics: Comte-Sponville

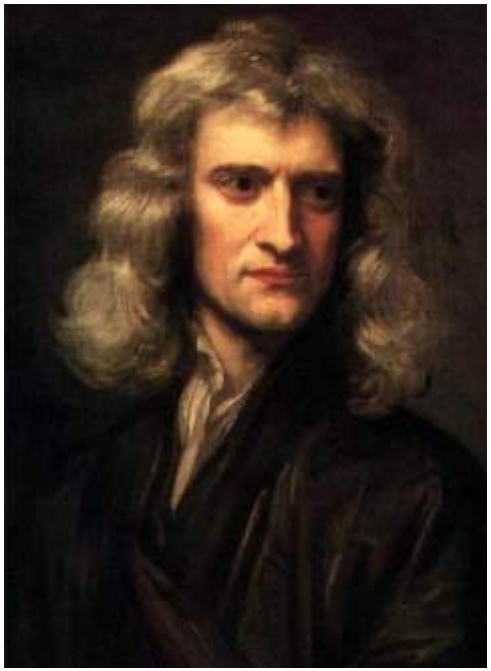


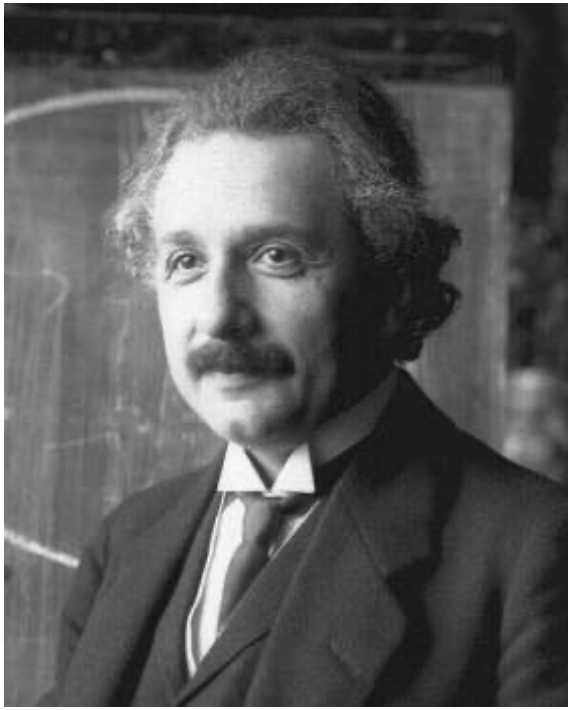
Moral: concerns Good and Evil, absolute or transcendental values ("what must I do to be just?")

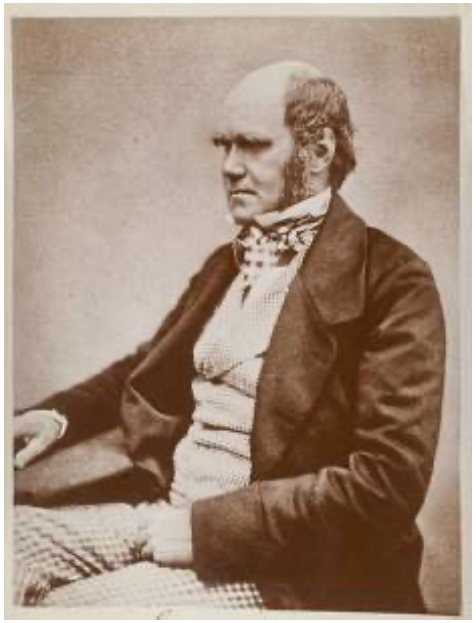
Ethics: deals with the good and the bad, values particular to society and the individual ("how to live to be happy?")

Ethics in Comte-Sponville: a set of practices of action of a community

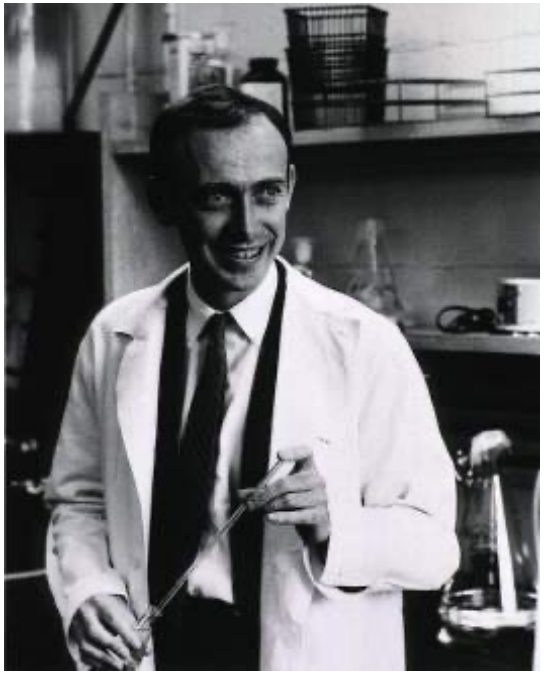
Ethics in Scientific Research



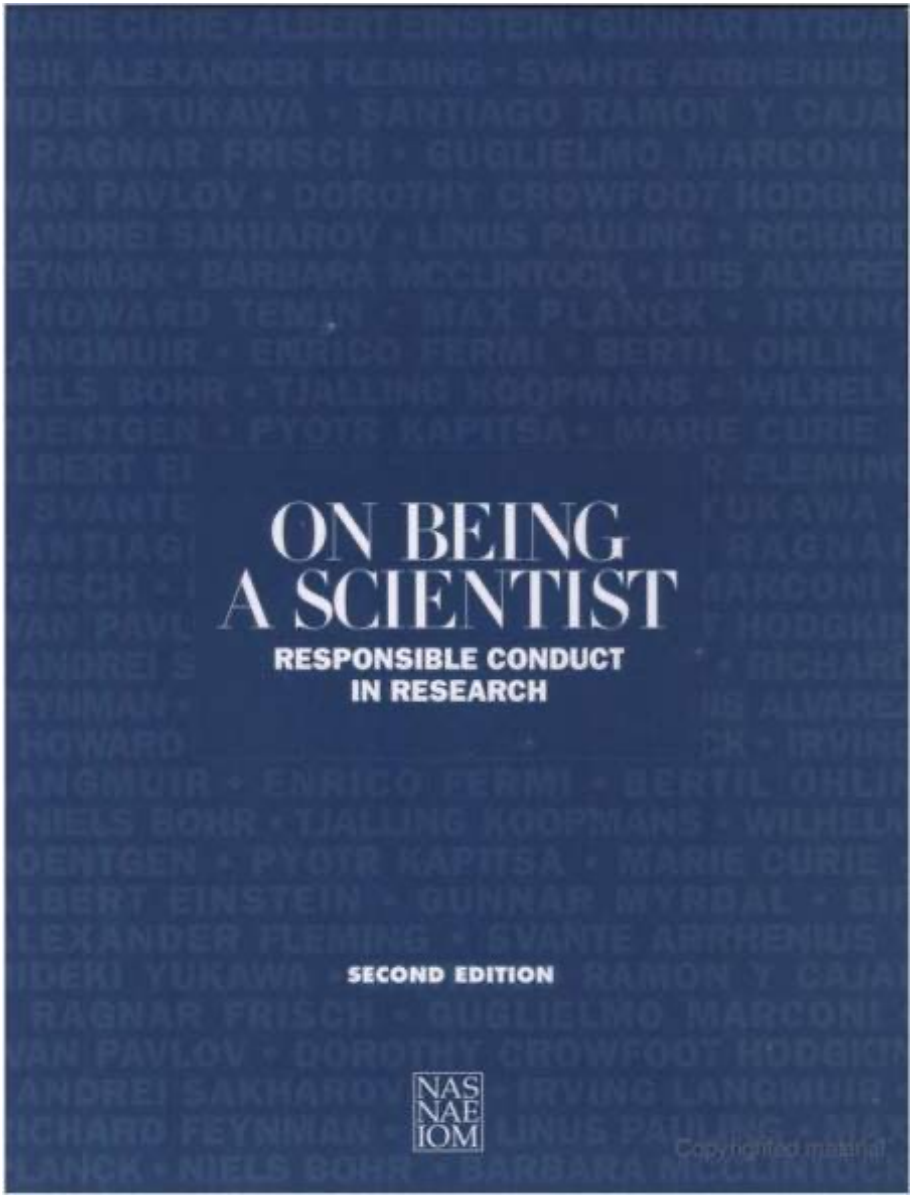




Set of practices adopted by the scientific community



Who defines accepted practices?



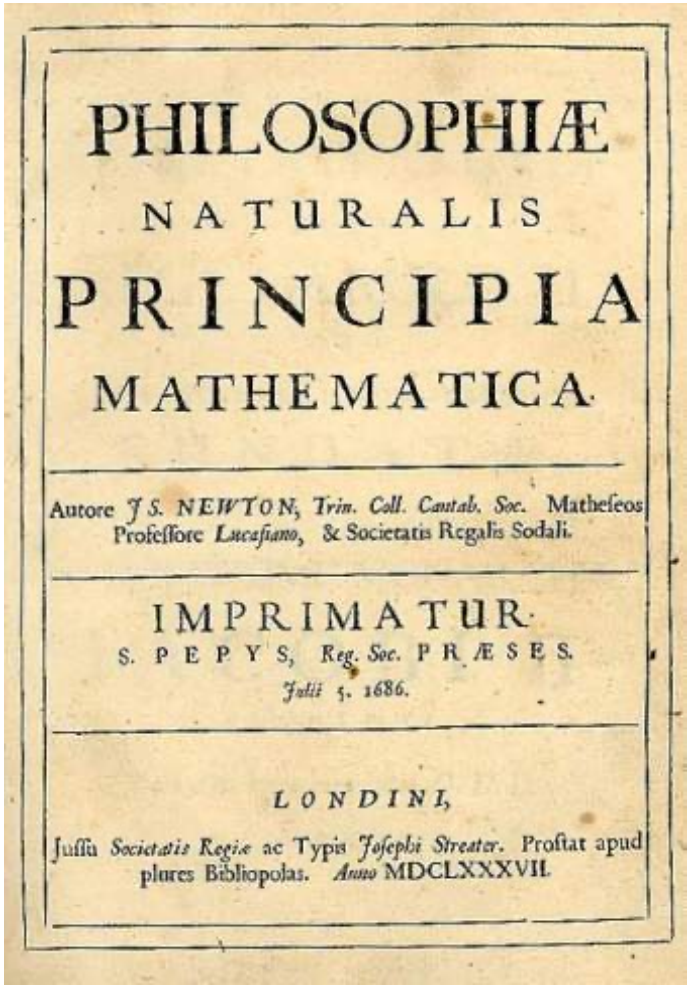
The scientific community is self-referential and self-regulated

Ethics of the scientific community

Primacy: whoever publishes the first article on the subject, has the credit for the discovery (or invention)

Advancement of knowledge: every scientific article must contain an unprecedented and relevant result for Science

Reproducibility: every published experiment must be replicable by others


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.

Autore JS. NEWTON, Trin. Coll. Cantab. Soc. Matheseos Professore Lucasiano, & Societatis Regalis Sodali.

IMPRIMATUR.
S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.

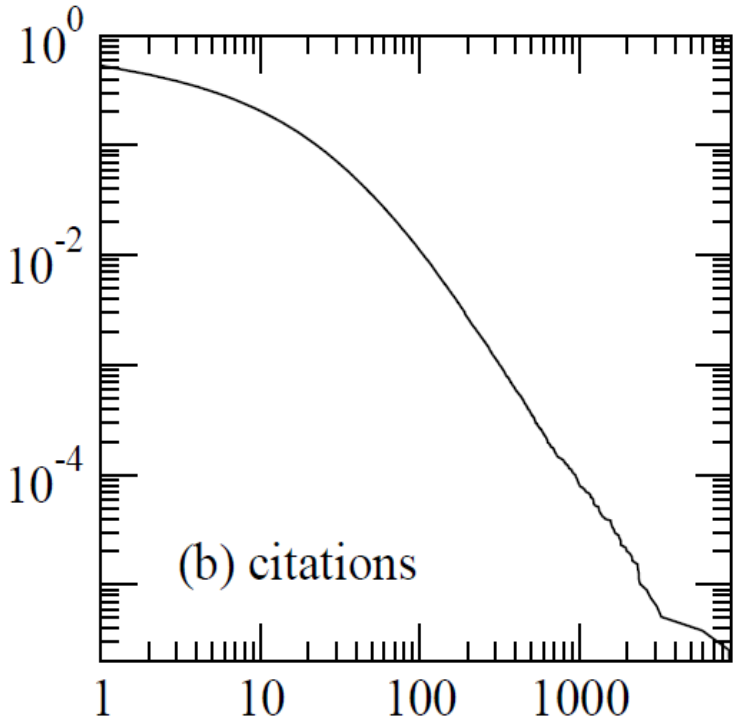
LONDINI,
Jussu Societatis Regiæ ac Typis Josephi Streater. Prostat apud plures Bibliopolas. Anno MDCLXXXVII.

How is Science measured?

Frequency: number of articles published per year

Impact: number of citations

Influence: ability to get money for research



"Matthew Effect"

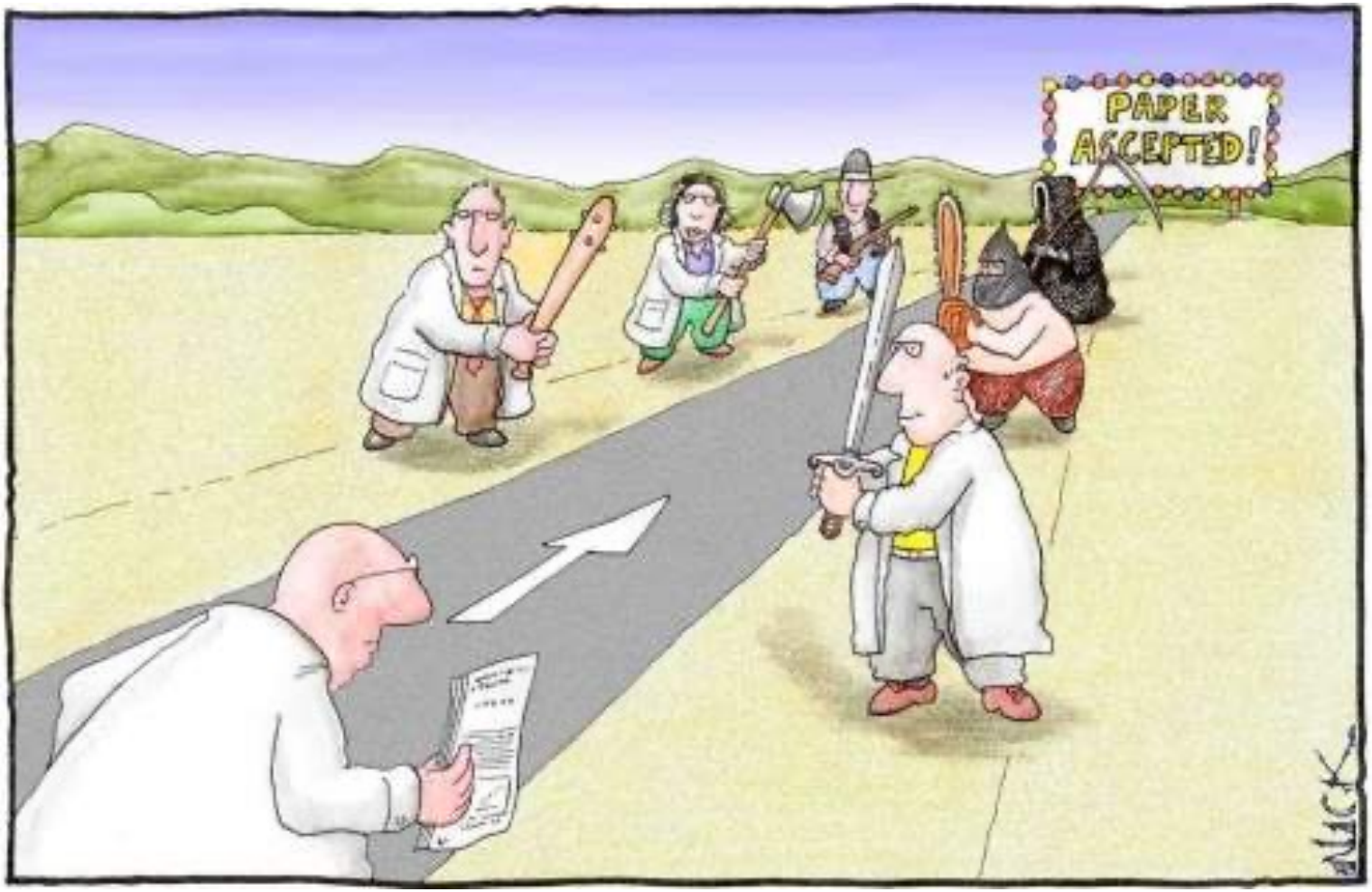
Prestige!

Publish or perish?

High-impact journals generate prestige...

Researchers live under pressure

Peer-review: the long and winding road



Peer review

Article received by the editor

↓

Submitted for external reviewers

↓

Reviewers recommendations

↓

New review

↓

Publication

↓

Some reviewers have bad manners

"The authors seem to be reinventing the wheel and a flat tire to go along with it."

"The study essentially collected some data and ran some correlations, so the reader learns nothing new."

Some reviewers have bad manners

"Finally, the paper is poorly and hastily written, with typos, poor organization, and poor characterization of the basic work it's built on."

"The arguments in the paper are compelling but not convincing."

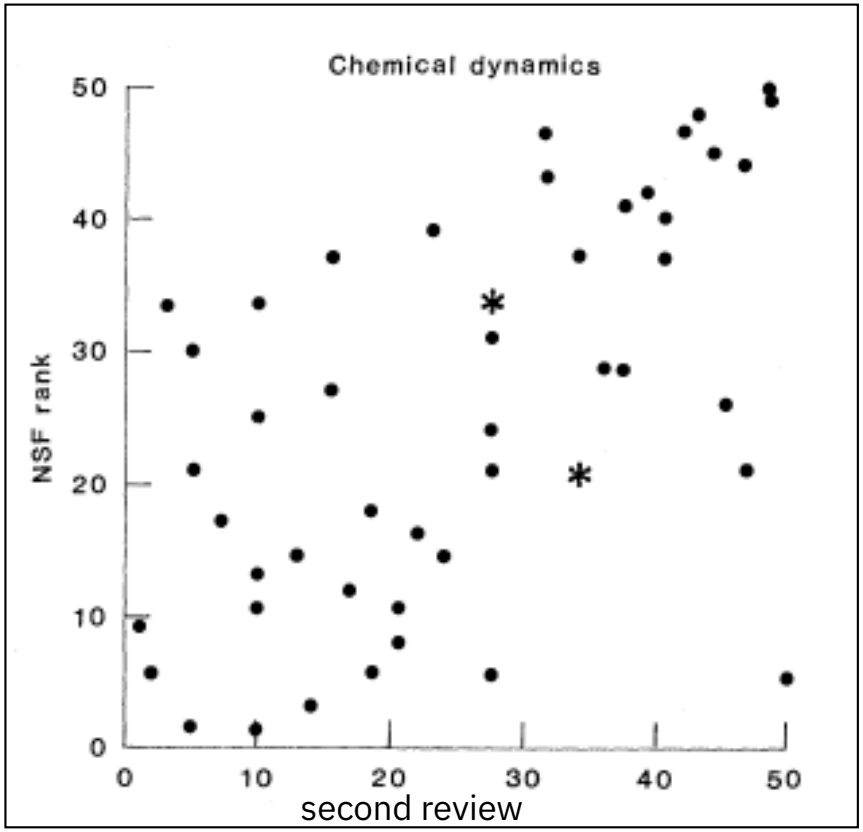
Problems with peer review

Image source: The Conversation

“Peer review is slow, expensive, profligate of academic time, highly subjective, prone to bias, easily abused, poor at detecting gross defects, and almost useless in detecting fraud.” R. Smith, BMJ

Subjectivity in peer review

www.jamesyang.com



50 applications for NSF funding (Chemistry)

Independently assessed twice

Conclusion: judgment depends on who reviews

S Cole et al, “Chance and consensus in peer review”, Science 1991

Every author's dream

www.jamesyang.com

Hyung-In Moon

"Moon suggested preferred reviewers during the submission which were him or colleagues under bogus identities and accounts. The reviews were almost always favourable but still provided suggestions for paper improvement."

source: retractionwatch

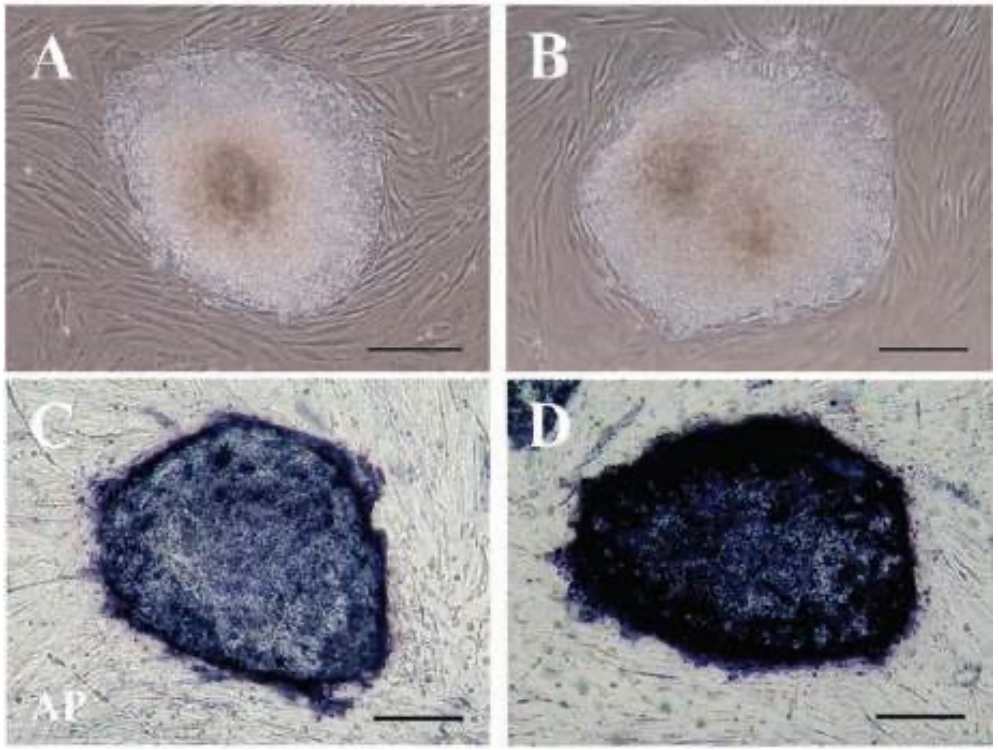
Hwang Woo-Suk and Human Cloning

Dog cloning
(2004)

Patient-Specific Embryonic Stem Cells Derived from Human SCNT Blastocysts

Woo Suk Hwang,[1,2*] Sung Il Roh,[3] Byeong Chun Lee,[1]
Sung Keun Kang,[1] Dae Kee Kwon,[1] Sue Kim,[1] Sun Jong Kim,[3]

Science, 3008, 1777 (2005)

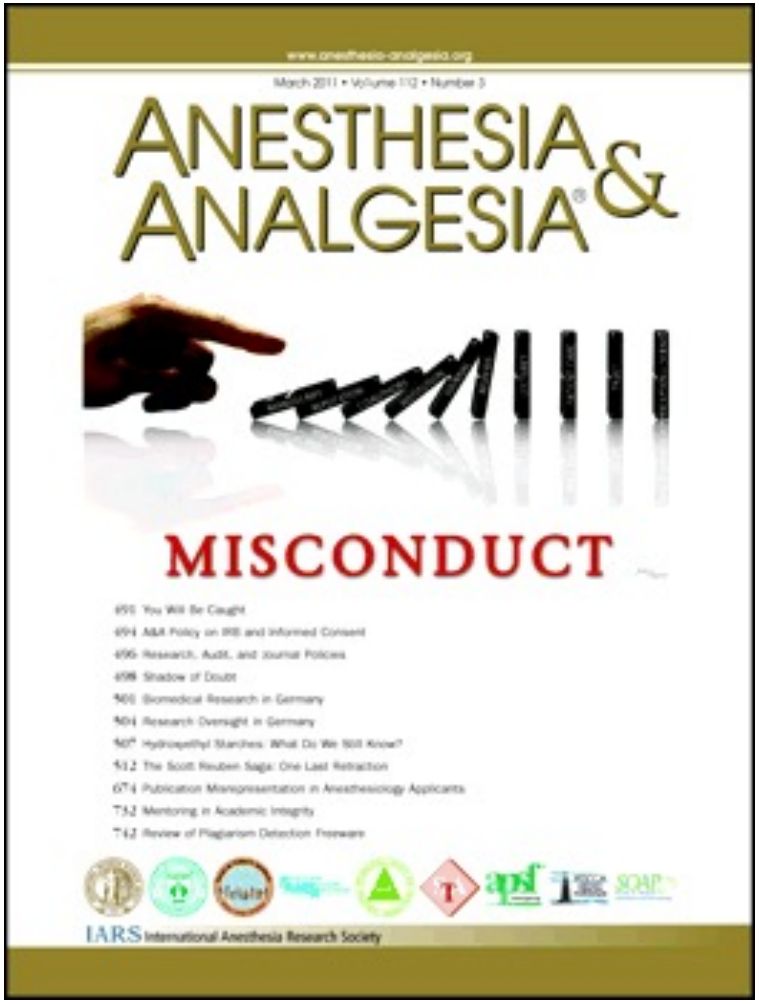


Hwang Woo-Suk and Human Cloning

Editorial Expression of Concern

On 23 December 2005, the Seoul National University Investigation Committee provided an interim report on their investigation of Woo Suk Hwang's research. The report (*2*) stated that "the experimental data submitted to *Science* in support of 11 stem cell lines (DNA fingerprinting, microscopic photos, confirmation of teratomas, etc.) were all derived from 2 cell lines" and that "the Committee finds that the experimental data published in the 2005 *Science* paper were based on a deliberate manipulation, in other words a fabrication of research results." The report also states that "The Investigation Committee has submitted samples of cell lines 2 and 3 for DNA testing in order to determine their authenticity."

Book of records: 172 papers canceled?



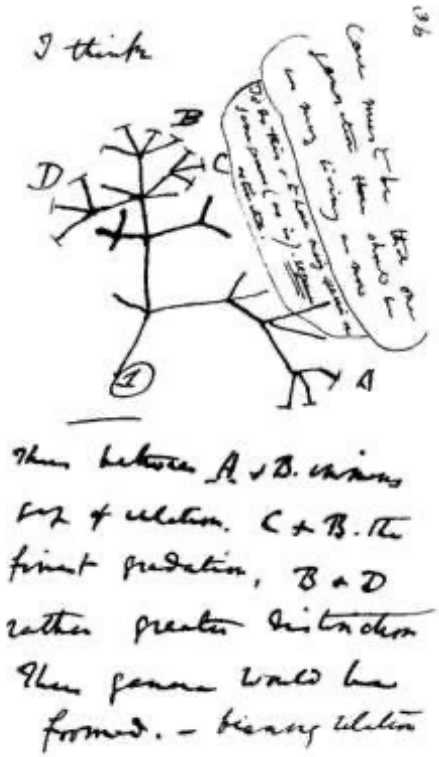
Dr. Yoshitaka Fujii (prof. Anestesiologia Univ. Tokyo)

The Francesca Gino affair

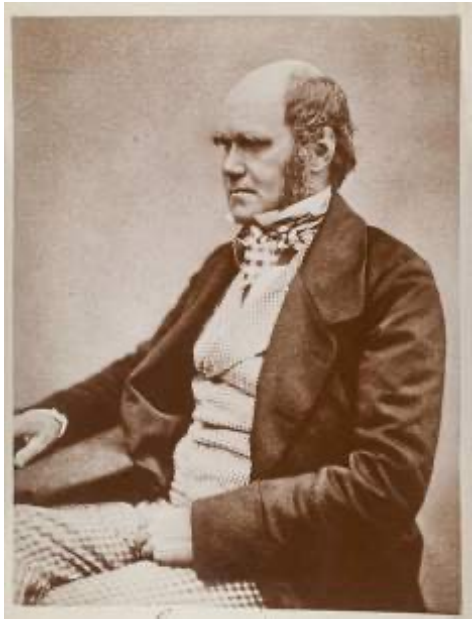
Gino published papers about honest behaviour.
There is strong evidence that she falsified data.

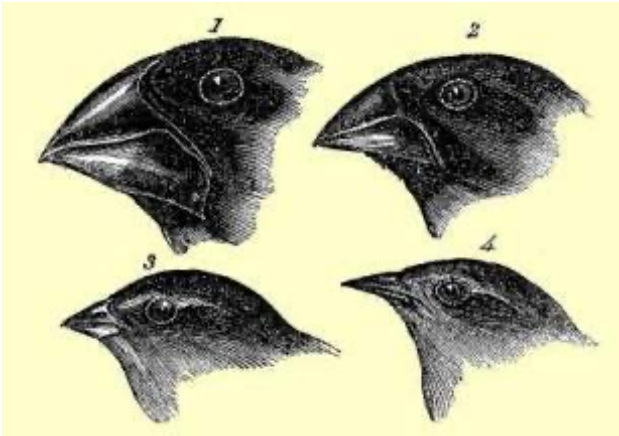
Darwin and the Origin of Species

Voyage of the Beagle (1831-1836)



Darwin’s notes

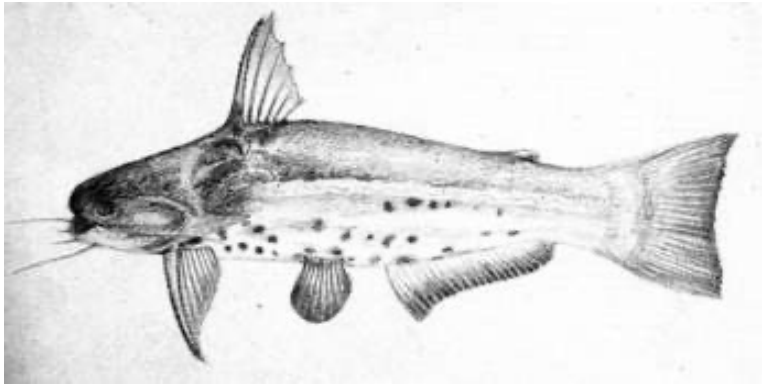




Galapagos finches

Wallace and natural selection

A.R. Wallace (1823-1913)

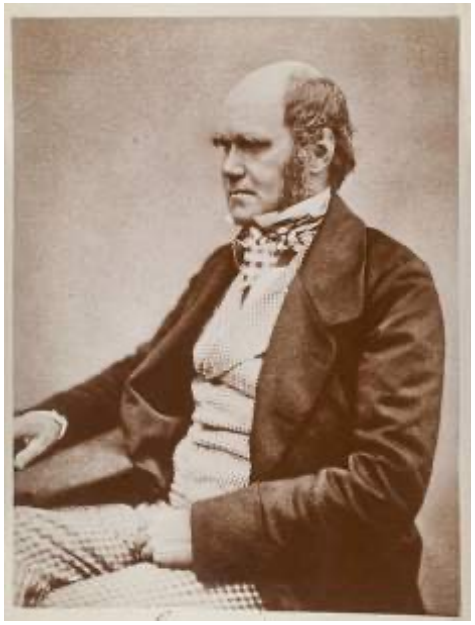


Fishes in the Amazon

Birds in Indonesia

Darwin, Wallace, and Natural Selection

"On The Tendency of Varieties to Depart Indefinitely from the Original Type"
(June, 1858)



The LINNEAN SOCIETY of London

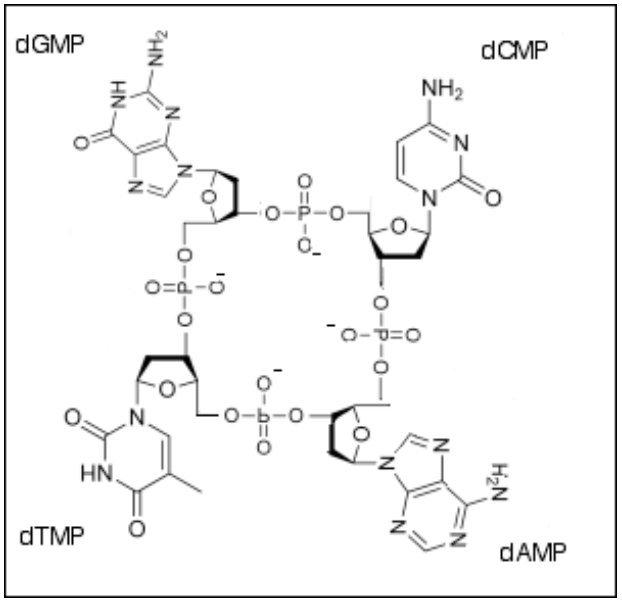
"All my originality will be smashed"
(letter to Lyell)

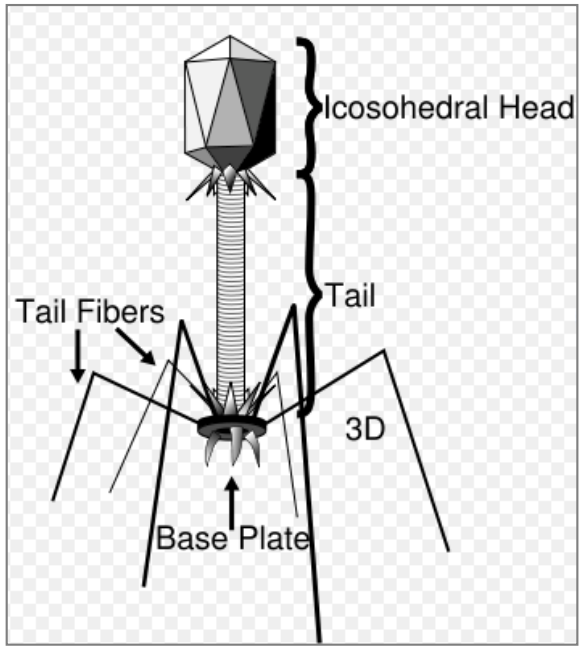
Charles Lyell, editor

DNA: Theory and observations

Meischer isolated DNA (1871)

Levene discovered composition (ATCG) but proposed the wrong structure (1919)





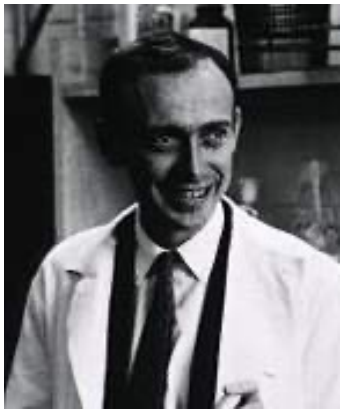
Hershey and Chase: DNA is the genetic material of the T2 virus (1952)

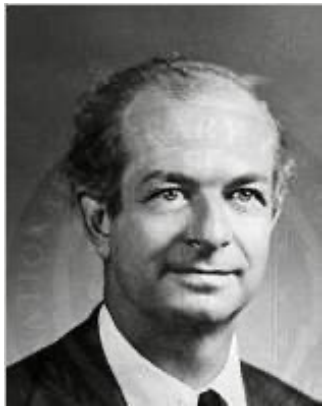
Who will discover the structure of DNA?

King´s College Londres
(Wilkings and Franklin)

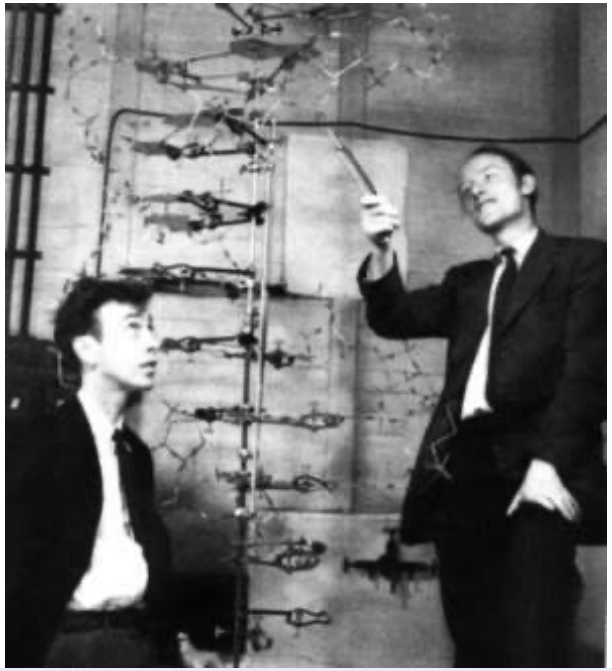
Cambridge
(Watson and Crick)

CALTECH
(Pauling)

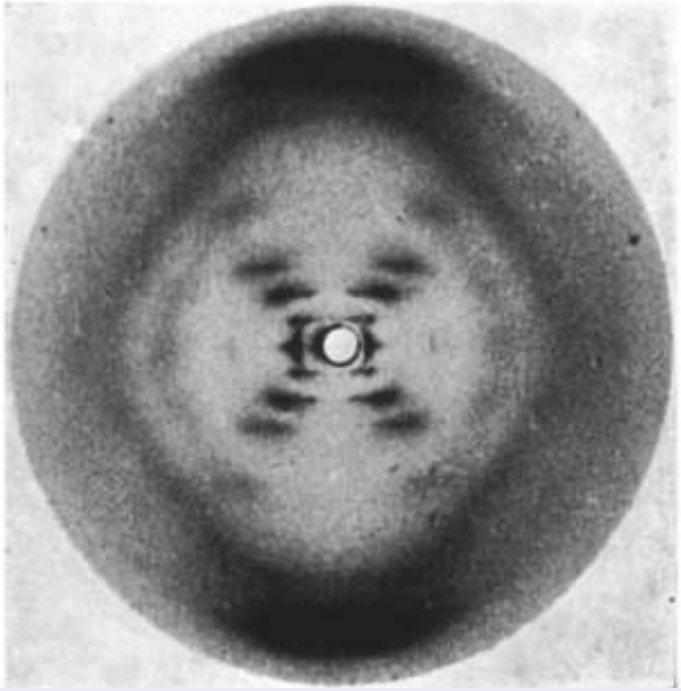




DNA: Theory and observations



Watson and Crick build models out of cardstock. No experimental data



Watson has access to "Photo 51" by Rosalind Franklin (1952)

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

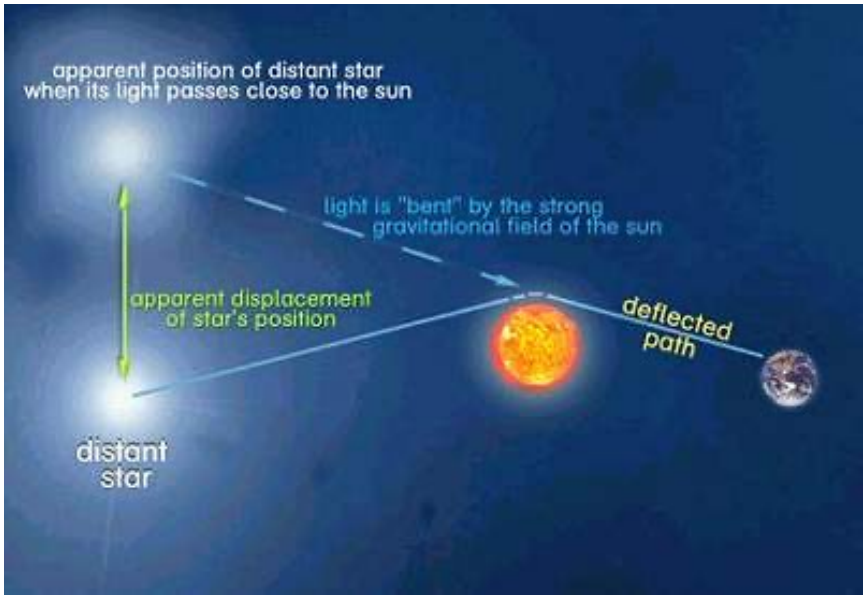
A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

Watson and Crick Nature, 171, pp 737-78 (1953)

Atomic DNA imaging

A Crucial Experiment: The 1919 Eclipse



Deviation of light predicted by Einstein

Einstein's prediction, made in 1916, that the light from a star should be deflected by 1.7" when passing very close to the Sun, was tested during a total eclipse that occurred on May 29, 1919. This eclipse was visible in a band that stretched from Brazil to Africa

A Crucial Experiment: The 1919 Eclipse

English expeditions sent to Sobral and Príncipe

Photo taken by Eddington in Principe to measure the deviation in the location of the stars

Expedition equipment in Sobral

A Crucial Experiment: The 1919 Eclipse

Problem: The data didn't match

Prediction of Newton's theory 0.87”

Einstein's theory prediction 1.75”

Sobral (instrument 1) 1.98”

Sobral (instrument 2) 0.93”

Principe (Instrument 3) 1.61"

What did Arthur Eddington (chief scientist) do?

A Crucial Experiment: The 1919 Eclipse

Eddington ignored the bad data from Sobral!

IX. *A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations made at the Total Eclipse of May 29, 1919.*

By Sir F. W. Dyson, *F.R.S., Astronomer Royal, Prof.* A. S. Eddington, *F.R.S., and Mr.* C. Davidson.

(*Communicated by the Joint Permanent Eclipse Committee.*)

Received October 30,—Read November 6, 1919.

[Plate 1.]

Contents.

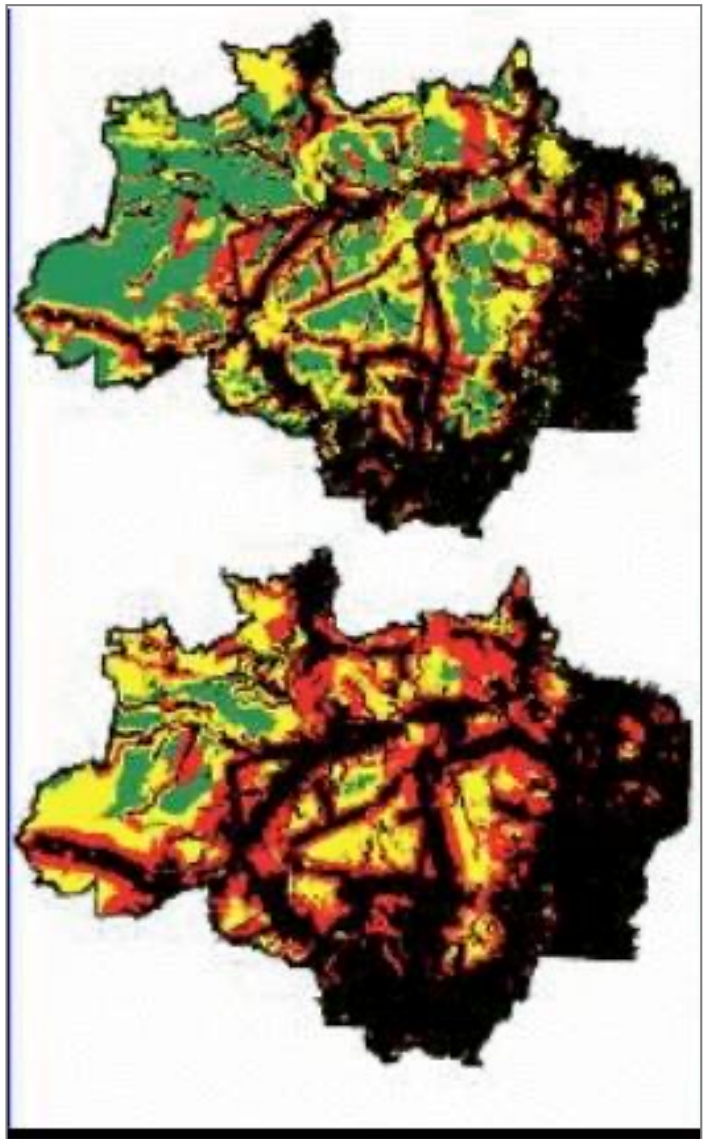
		Page
I.	Purpose of the Expeditions	291
II.	Preparations for the Expeditions	293
III.	The Expedition to Sobral	296
IV.	The Expedition to Principe	312
V.	General Conclusions	330

It was only in 1979 that Eddington's data were reanalyzed and the error in Sobral's measurement was described

Harvey, G. M. (1979). "Gravitational Deflection of Light: A Re-examination of the observations of the solar eclipse of 1919." The Observatory 99, 195-198.

Is it honest to exaggerate to publish?

W. Laurance et al, “The Future of the Brazilian Amazon?”, Science, 2001



Scenarios for the Amazon in 2020 according to Laurance et al, 2001

Optimistic scenario: 28% deforestation

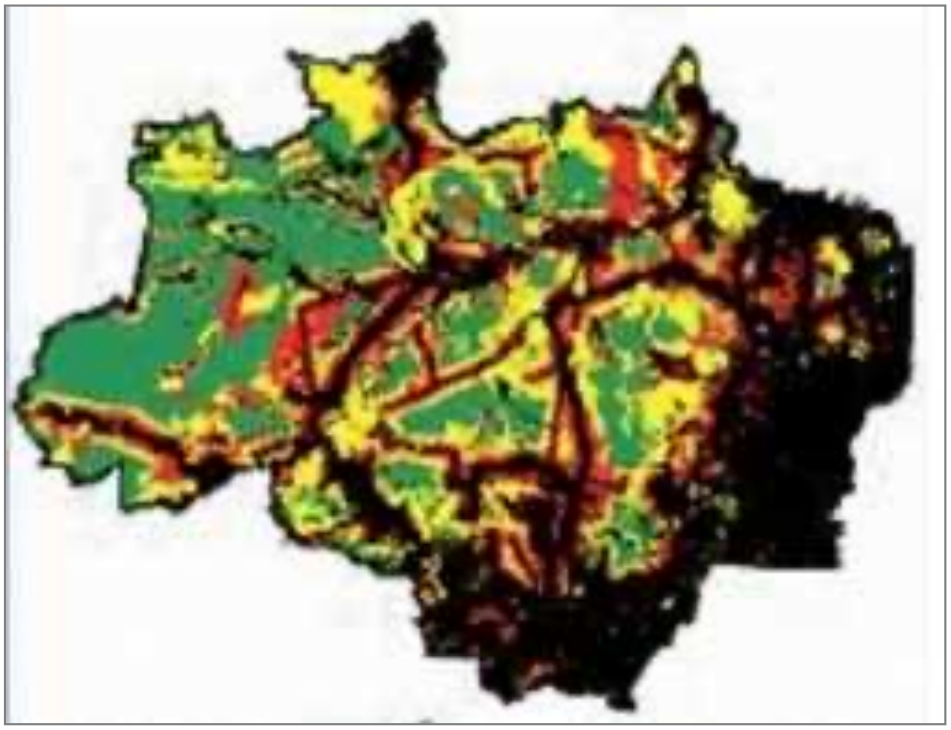
Pessimistic scenario: 42% deforestation

“We generated two models with realistic but differing assumptions for the future of the Brazilian Amazon. The models predict the spatial distribution of deforested or heavily degraded land, as well as moderately degraded, lightly degraded, and pristine forests”.

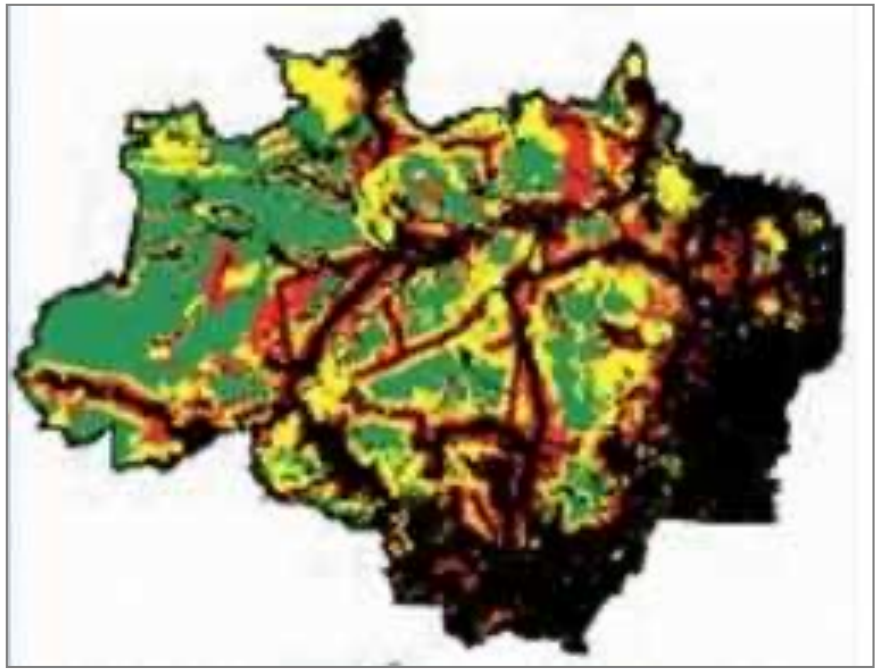
The Future of Brazilian Amazonia?

Optimistic scenario: 28% of deforestation (1 million km^2) by 2020

- Complete degradation up to 20 km from roads (existing and projected)
- Moderate degradation up to 50 km from roads
- Reduced degradation up to 100 km from roads

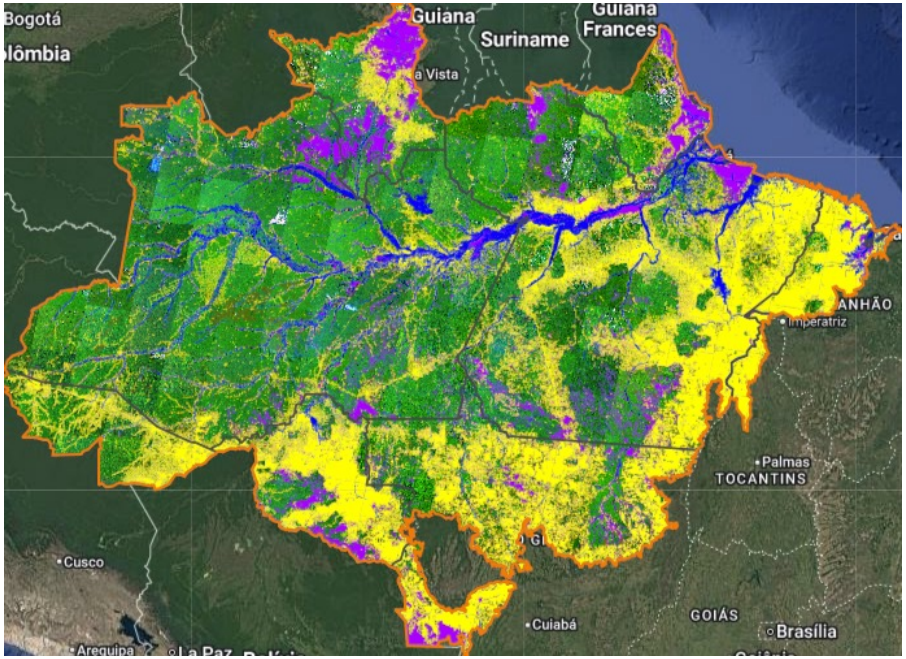


Disaster scenario and observed reality...



Laurance et al., 2001
Optimistic scenario (2020)

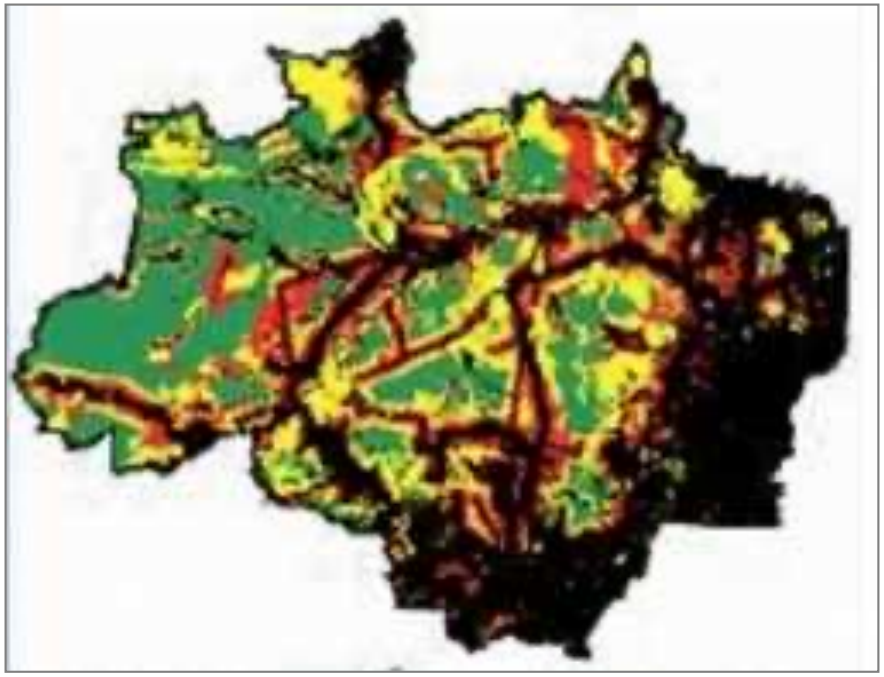
- Savannas and deforestation
- Moderate degradation
- Mild degradation
- Pristine forest



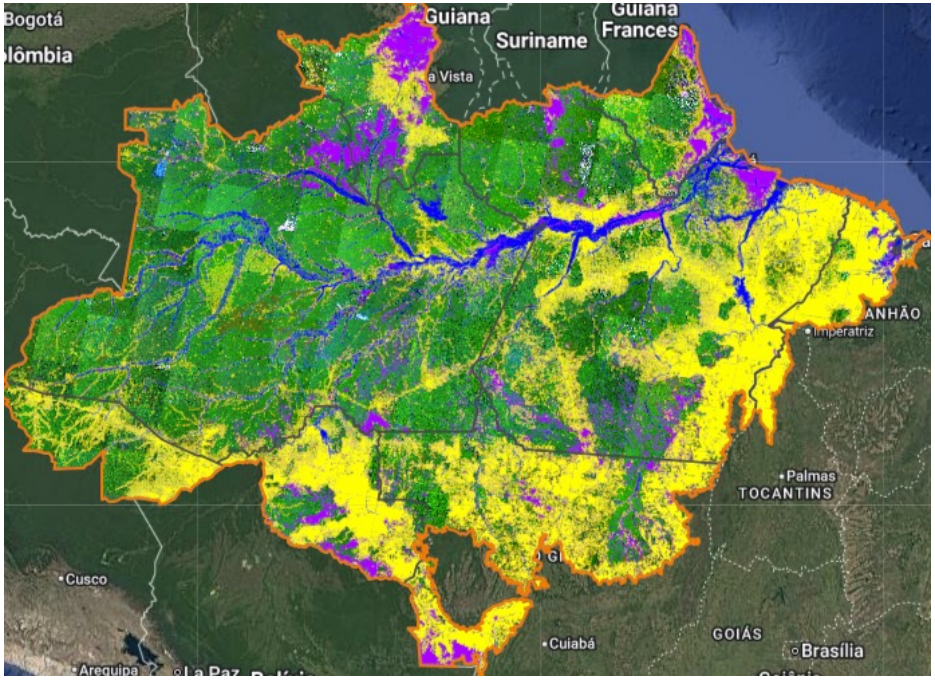
Data from INPE (Prodes, 2025)

- Savannas
- Deforestation
- Forest

Disaster scenario and observed reality...



Laurance et al., 2001
Optimistic scenario (2020)



Data from INPE (Prodes, 2025)

For Laurence's "optimistic" scenario to happen, Brazil would have to deforest 50 thousand km^2 per year from 2010 to 2020!

Prejudice against researchers from the South?

How to publish without being part of the main research centre?

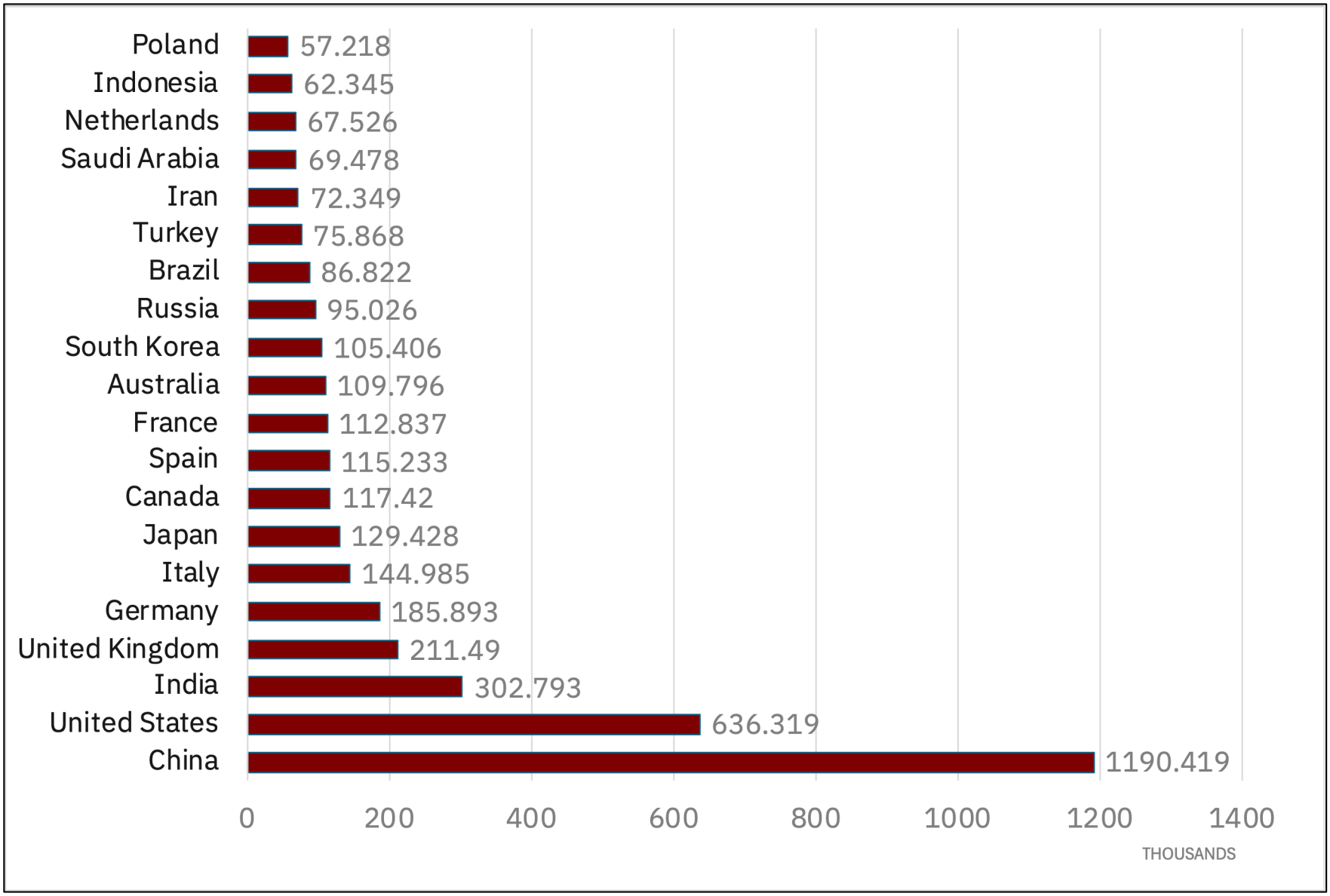
What would you do?

The authors need to explicitly state that the conceptual contributions of this manuscript are incremental and somewhat marginal: the kernel estimation method is just one of many specific ways to derive the composite population count, and local segregation measures were derived using this specific derivation of composite population count. In fact, it is not clear how the kernel estimation approach is better as compared to, say the distance decay approach used in Wong (2005). The randomization test is another contribution in the context of testing the significance

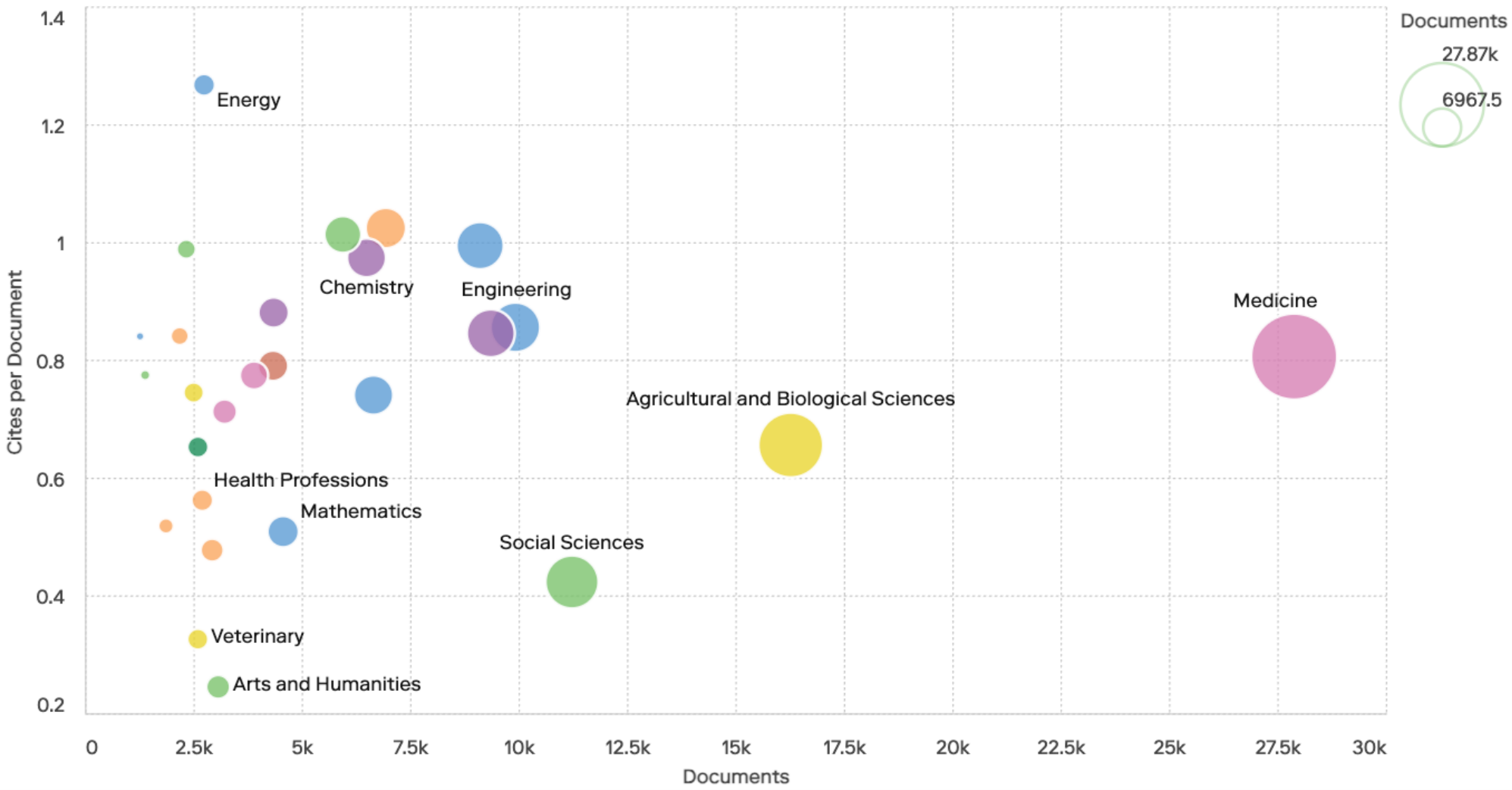
F Feitosa, G Câmara, A Monteiro, T Koschitzki, M Silva, “Global and Local Spatial Indices of Urban Segregation”, International Journal of Geographical Information Science, vol 21(3):299-323, March 2007.

Countries ranked by papers (2024)

source: SCImago



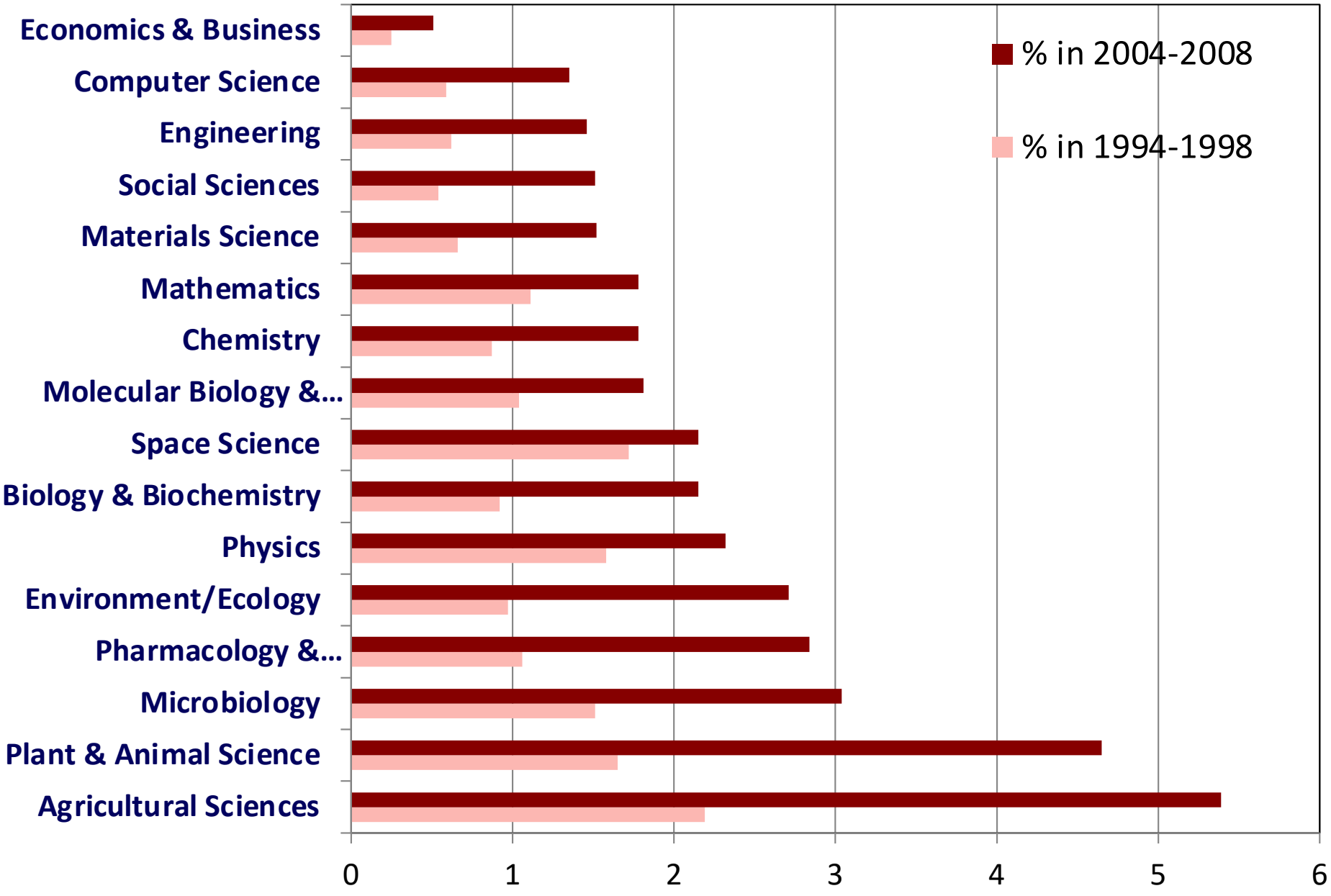
Brazil’s scientific production (2024-2025)



source: SCImago

Science in Brazil (% of world)

source: ISI



Brazilian science (2003-2007)

TABLE 2
Brazil's share of world output in ten fields in *Web of Science*

Field	Share (% of world)	Volume (papers 2003-07)
Tropical Medicine	18.40	1,433
Parasitology	12.34	1,635
Multidisciplinary Agriculture	8.61	1,627
Oral Surgery & Medicine	8.19	2,203
Entomology	7.06	1,629
Dairy & Animal Sciences	6.49	1,617
Biology	6.43	1,999
Soil Sciences	5.84	947
Veterinary Sciences	5.79	3,421
Zoology	5.57	2,264

In some subareas of Life Sciences, Brazil is one of the world's leaders

source: "Global Research Report - Brazil ", Thomson Reuters, 2009

source: “Building BRICKS”, Thomson Reuters, 2013

Brazil and other BRICKS (2007-2011)

BRAZIL (2.6)		RUSSIA (2.4)		INDIA (3.4)		CHINA (11)		S. KOREA (3.3)	
Agricultural Sciences	8.8	Physics	7.3	Chemistry	6.4	Materials Science	24.5	Materials Science	6.3
Plant & Animal Science	6.6	Space Science	6.8	Pharmacology & Toxicology	6.1	Chemistry	20.2	Computer Science	5.6
Pharmacology & Toxicology	3.7	Geosciences	6.6	Agricultural Sciences	6.1	Physics	17.9	Engineering	5.1
Microbiology	3.3	Mathematics	4.7	Materials Science	5.9	Mathematics	15.7	Pharmacology & Toxicology	4.8
Environment / Ecology	3.0	Chemistry	4.5	Microbiology	5.1	Engineering	14.8	Physics	4.7
Social Sciences	2.8	Materials Science	3.1	Physics	4.3	Computer Science	13.1	Microbiology	4.2
Clinical Medicine	2.6	Engineering	2.1	Engineering	4.1	Geosciences	12.3	Chemistry	3.7
Biology & Biochemistry	2.6	Molecular Biology	2.0	Plant & Animal Science	4.0	Pharmacology & Toxicology	10.1	Agricultural Sciences	3.4
Neurosciences	2.6	Microbiology	1.7	Geosciences	3.7	Environment/ Ecology	9.8	Biology & Biochemistry	3.3
Immunology	2.5	Biology & Biochemistry	1.6	Biology & Biochemistry	3.6	Biology & Biochemistry	8.8	Clinical Medicine	2.8

Brazil is strong in Life Sciences

Science in Brazil: production and impact (2011-2016)

Categories	Papers	Citation Impact	Top 1%	Top 10%	% International Collaborations
Clinical Medicine	46,840	0.87	0.96	5.43	28.8
Plant & Animal Science	30,980	0.66	0.44	4.11	26.9
Agricultural Sciences	25,629	0.59	0.44	4.01	14.0
Chemistry	19,345	0.64	0.08	3.61	30.2
Physics	15,040	1.24	1.70	11.33	55.2
Engineering	12,092	0.85	0.45	6.99	32.2
Social Sciences, general	11,555	0.67	0.62	4.07	21.6
Biology & Biochemistry	11,537	0.69	0.38	4.50	32.6
Environment/Ecology	9,831	0.89	1.11	7.38	40.3
Materials Science	7,528	0.60	0.17	3.49	35.0
Pharmacology & Toxicology	7,456	0.79	0.46	5.10	25.8
Neuroscience & Behavior	7,176	0.79	0.56	5.56	36.8
Molecular Biology & Genetics	6,359	0.64	0.63	4.07	39.3
Mathematics	5,600	0.95	0.54	8.27	46.5
Immunology	5,171	0.88	0.87	6.61	44.4
Microbiology	5,160	0.76	0.58	5.08	37.9
Geosciences	5,042	0.82	0.69	5.91	53.0
Psychiatry/Psychology	3,576	0.91	1.37	7.35	45.6
Computer Science	3,560	0.85	0.56	6.88	43.2
Space Science	2,461	1.37	2.60	10.93	77.7
Economics & Business	1,897	0.58	0.26	4.43	36.0

source: “Brazil Research Report”, Thomson Reuters, 2016

UNESCO SCIENCE REPORT

The race against time for smarter development

Chapter 8: BRAZIL

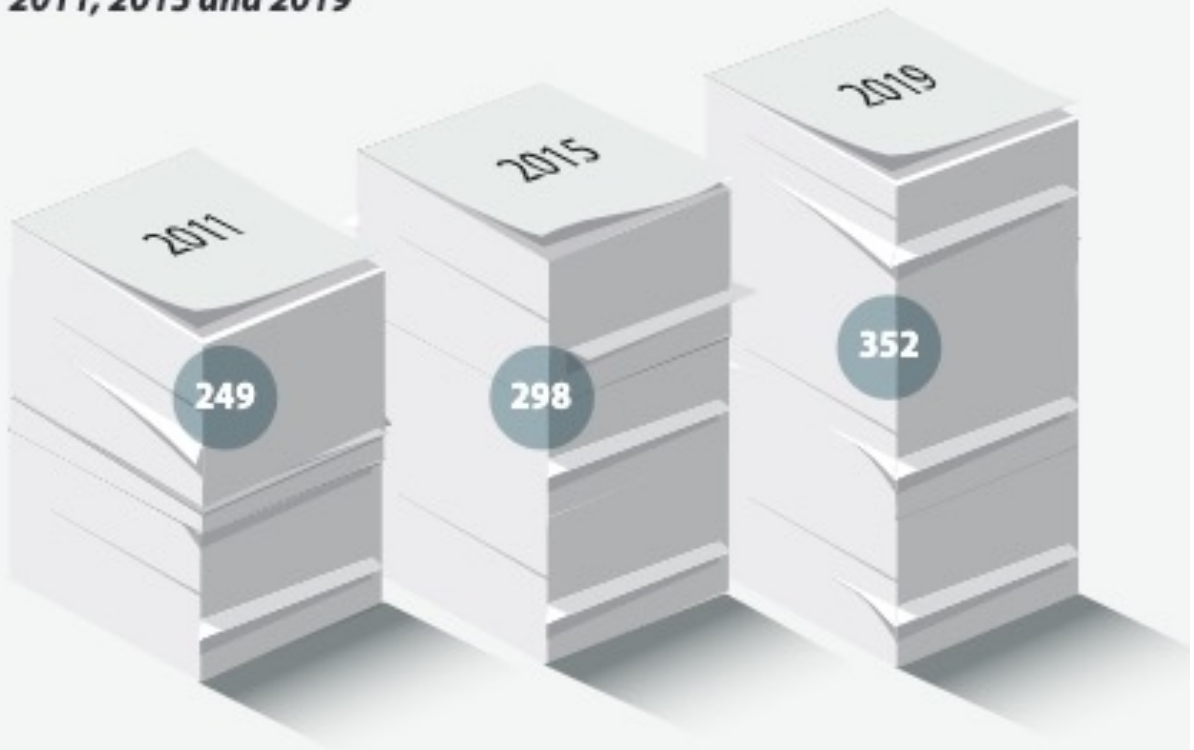
0.91

Average of relative citations for Brazil, 2014–2016; the G20 average is 1.02

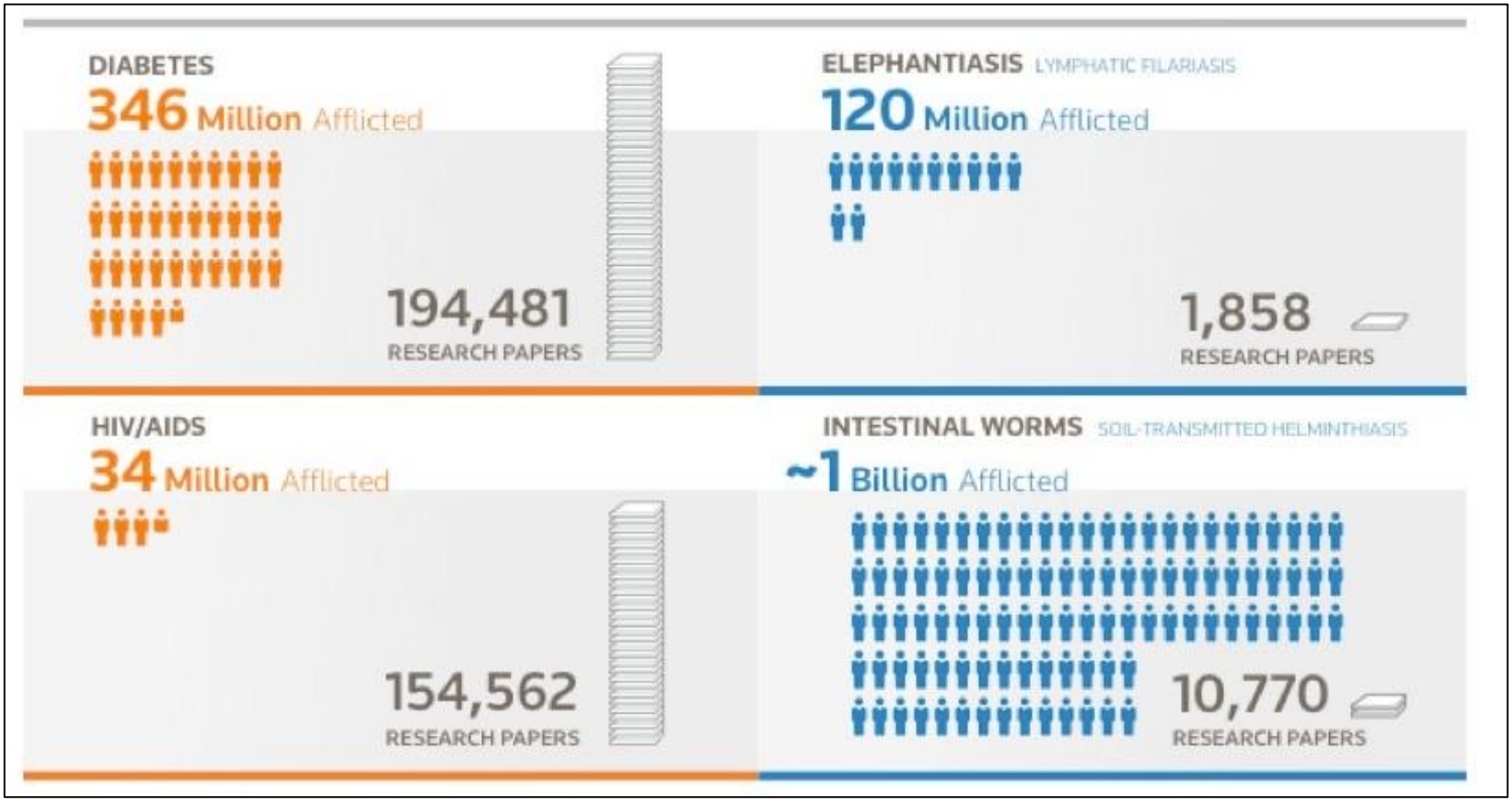
34%

Share of publications with foreign co-authors in Brazil, 2017–2019; the G20 average is 25%

Scientific publications per million inhabitants in Brazil, 2011, 2015 and 2019



The impact of neglected diseases



Poor man’s diseases are not hot scientific topics!

source: ISI

How has output on SDG-related topics evolved since 2012?

Brazilian scientists are publishing at least three times more than the global average proportion on the following topics: traditional knowledge, help for smallholder food producers, the sustainable use of terrestrial ecosystems and the status of terrestrial biodiversity. Brazil's output on agro-ecology is even 4.5 times the global average.

In health research, the intensity of output is 4.2 times the global average for tropical communicable diseases. Moreover, the number of publications on new or re-emerging viruses that can infect humans increased by 250% from 643 (2012–2015) to 1 605 (2016–2019), in the context of the Zika epidemic. Despite this scientific expertise, the WHO Covid-19 Dashboard reported in October 2020 that Brazil had the third-highest national caseload of this coronaviral disease after the USA and India.

Advice for young scientists from the South

Work with international collaborators

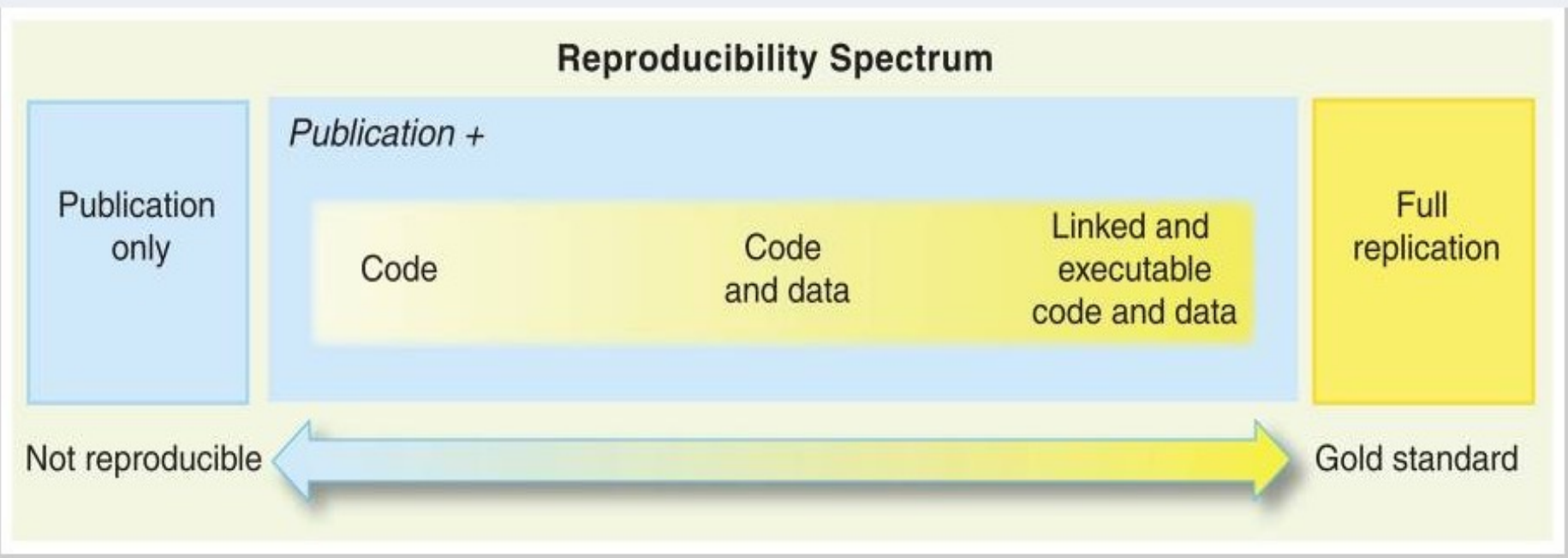
Write your articles with the best possible quality and don't expect to have a big impact

How to be happy as a Brazilian scientist?

The bigger your internal agenda can be, the greater your contribution to the Brazilian economy

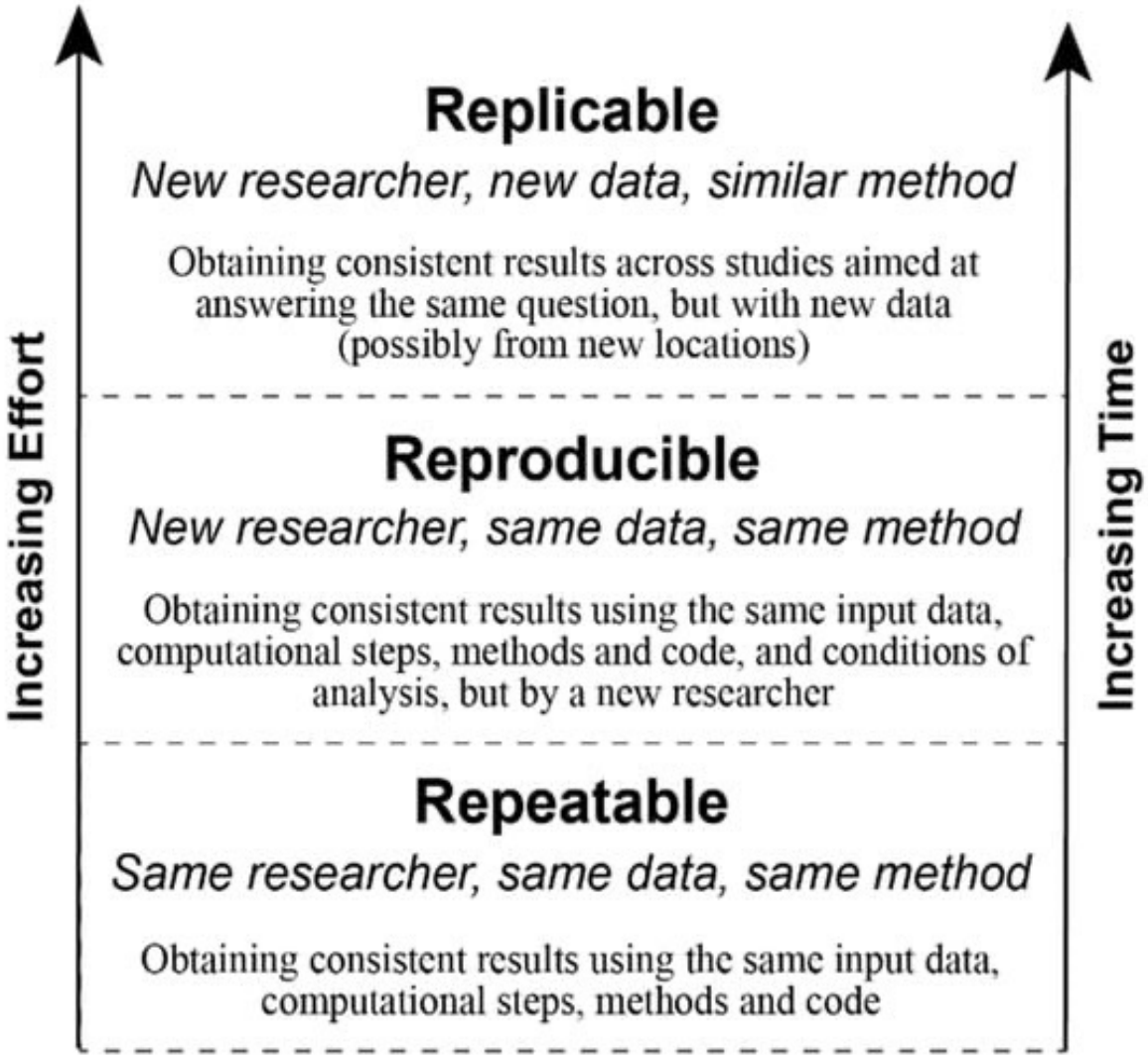
Or would you like to be one of them?

Achieving reproducible research



Peng RD, “Reproducible research in computational science”. Science, 2011

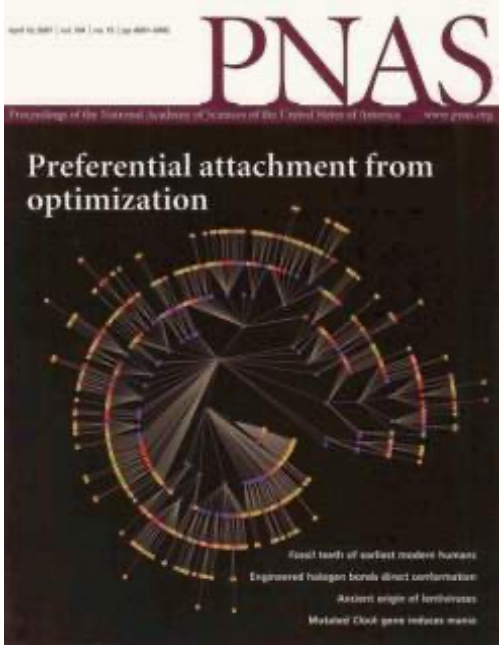
The road to Open Science



Kedron & Frazier, “How to Improve the Reproducibility, Replicability, and Extensibility of Remote Sensing Research”, Remote Sensing, 2021.

Scientific ethics is amoral

Follow the rules of scientific ethics to be able to publish



Have no illusions:

1. Science is a competitive activity
2. The limits of scientific ethics are fluid
3. There are a lot of dishonest people
4. Try to keep your integrity anyway!